(iii) $(5x+4y)(5x-4y)$ $\left[\text{we have } (a+b)(a-b) = a^2-b^2\right]$

$(5x+4y)(5x-4y) = (5x)^2-(4y)^2$ put $[a=5x \quad b=4y]$

$= 25x^2-16y^2$

(iv) $(m+5)(m-8)$ $\left[\text{we have } (x+a)(x-b) = x^2+(a-b)x-ab\right]$

$(m+5)(m-8) = m^2+(5-8)m-(5)(8)$ put $[x=m,\ a=5,\ b=8]$

$= m^2-3m-40$

3.4.1 Expansion of Trinomial $(a+b+c)^2$

We know that $(x+y)^2 = x^2+2xy+y^2$

Put $x=a+b,\ y=c$

Then, $(a+b+c)^2 = (a+b)^2+2(a+b)(c)+c^2$

$= a^2+2ab+b^2+2ac+2bc+c^2$

$= a^2+b^2+c^2+2ab+2bc+2ca$

Thus, $(a+b+c)^2 \equiv a^2+b^2+c^2+2ab+2bc+2ca$

$(a+b+c)^2$

	a	b	c
a	a^2	ab	ca
b	ab	b^2	bc
c	ca	bc	c^2

Example 3.17 Expand $(a-b+c)^2$

Solution

Replacing 'b' by '$-b$' in the expansion of

$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$

$(a+(-b)+c)^2 = a^2+(-b)^2+c^2+2a(-b)+2(-b)c+2ca$

$= a^2+b^2+c^2-2ab-2bc+2ca$

Progress Check

Expand the following and verify :

$(a+b+c)^2 = (-a-b-c)^2$

$(-a+b+c)^2 = (a-b-c)^2$

$(a-b+c)^2 = (-a+b-c)^2$

$(a+b-c)^2 = (-a-b+c)^2$

Example 3.18 Expand $(2x+3y+4z)^2$

Solution

We know that,

$(a+b+c)^2 = a^2+b^2+c^2+2ab+2bc+2ca$

Substituting, $a=2x, b=3y$ and $c=4z$

$(2x+3y+4z)^2 = (2x)^2+(3y)^2+(4z)^2+2(2x)(3y)+2(3y)(4z)+2(4z)(2x)$

$= 4x^2+9y^2+16z^2+12xy+24yz+16xz$

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